ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 2 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Section A: objective questions (30) · Section B: magnetism and electromagnetism, transformers, DC power supplies, three-phase systems, capacitors in series and parallel, DC machines, and an introduction to PLCs (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The unit of magnetic flux is the …

A tesla. B henry. C weber. D farad.

1.1.2 When a conductor cuts magnetic lines of force, an emf is induced. This is …

A Ohm's law. B Kirchhoff's law.
C Faraday's law. D Coulomb's law.

1.1.3 The direction of an induced current always opposes the change that caused it. This is …

A Lenz's law. B Fleming's left-hand rule only.
C Joule's law. D Newton's third law.

1.1.4 A transformer with more turns on the secondary than on the primary is a …

A step-down transformer. B isolation transformer only.
C rectifier. D step-up transformer.

1.1.5 Transformers only work on …

A DC. B AC.
C batteries. D solar panels.

1.1.6 The component in a power supply that changes AC into pulsating DC is the …

A capacitor. B transformer.
C fuse. D rectifier.

1.1.7 A 7805 integrated circuit is a …

A timer. B 5 V voltage regulator.
C logic gate. D motor driver.

1.1.8 In a star-connected three-phase system, the line voltage is … times the phase voltage.

A 2 B √2 C √3 D 3

1.1.9 Capacitors connected in parallel …

A add their capacitances. B have a smaller total capacitance.
C must all have different voltages. D block DC less.

1.1.10 The part of a DC motor that reverses the current in the armature coils is the …

A field winding. B commutator.
C bearing. D yoke.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Like magnetic poles attract each other.

1.2.2 Soft iron is used in electromagnets because it loses its magnetism easily when the current stops.

1.2.3 Increasing the number of turns on a coil increases the strength of its magnetic field.

1.2.4 A full-wave rectifier uses only half of each AC cycle.

1.2.5 A larger smoothing capacitor gives less ripple.

1.2.6 Three-phase supplies are used to drive large industrial motors.

1.2.7 Capacitors in series have a total capacitance larger than the largest capacitor.

1.2.8 A DC generator converts mechanical energy into electrical energy.

1.2.9 A PLC is a computer designed for controlling machines in harsh industrial environments.

1.2.10 Ladder logic is a programming language used in PLCs.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Region around a magnet where its force acts	A	core
1.3.2	Coil of wire that becomes a magnet when current flows	B	bridge rectifier
1.3.3	Rule that gives the direction of force on a current-carrying conductor (motor)	C	magnetic field
1.3.4	Rule that gives the direction of induced current (generator)	D	Fleming's right-hand rule
1.3.5	Laminated iron that carries the flux in a transformer	E	relay
1.3.6	Small AC variation left on a DC output	F	Fleming's left-hand rule
1.3.7	Four diodes connected to give full-wave rectification	G	armature
1.3.8	Rotating part of a DC motor that carries the coils	H	solenoid

1.3.9	Electrically operated switch with a coil and contacts	I	input module
1.3.10	PLC section that receives signals from switches and sensors	J	ripple

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: MAGNETISM AND ELECTROMAGNETISM

2.1 Sketch the magnetic field between the north pole of one bar magnet and the south pole of another bar magnet placed facing each other. (3)

2.2 State THREE ways to increase the strength of an electromagnet. (3)

2.3 Explain how a relay uses electromagnetism to let a small Arduino current switch a large motor current. (4)

2.4 State Faraday's law and name THREE factors that determine the size of the induced emf. (5)

[15]

QUESTION 3: TRANSFORMERS AND POWER SUPPLIES

3.1 A transformer has 1 150 primary turns and 60 secondary turns. The primary is connected to 230 V AC.

3.1.1 Is this a step-up or step-down transformer? (1)

3.1.2 Calculate the secondary voltage. (3)

3.1.3 If the secondary current is 2 A, calculate the primary current (assume an ideal transformer). (3)

3.2 Draw a block diagram of a regulated 5 V DC power supply for a robot, and explain the function of EACH block. (10)

3.3 Sketch the output waveforms of a half-wave rectifier and a full-wave rectifier. (4)

[21]

QUESTION 4: THREE-PHASE AND CAPACITORS

4.1 A star-connected three-phase supply has a phase voltage of 230 V. Calculate the line voltage. (2)

4.2 Give TWO advantages of three-phase supply over single-phase supply. (2)

4.3 Three capacitors of 10 µF, 20 µF and 20 µF are available.

4.3.1 Calculate the total capacitance if all three are connected in parallel. (2)

4.3.2 Calculate the total capacitance if all three are connected in series. (4)

4.3.3 Calculate the charge stored on the parallel combination when it is connected to 12 V. (2)

[12]

QUESTION 5: DC MACHINES AND PLCs

5.1 Name FOUR main parts of a DC motor and the function of each. (8)

5.2 Explain the difference between a DC motor and a DC generator in terms of energy conversion. (2)

5.3 Name the FOUR main parts of a PLC. (4)

5.4 A conveyor motor must start when a START button is pressed and keep running until a STOP button is pressed. Draw a simple ladder diagram with a latching (seal-in) contact for this. (6)

5.5 Give TWO advantages of using a PLC instead of hard-wired relay control. (2)

[22]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER